



Osteoarthritis

Addis Ababa University

By

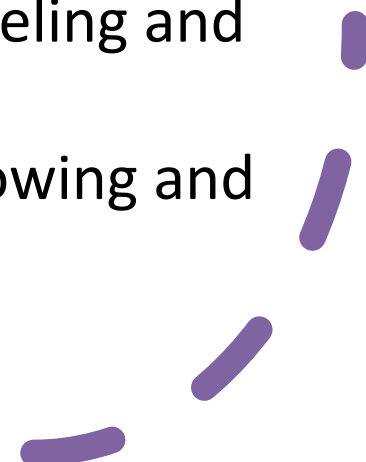
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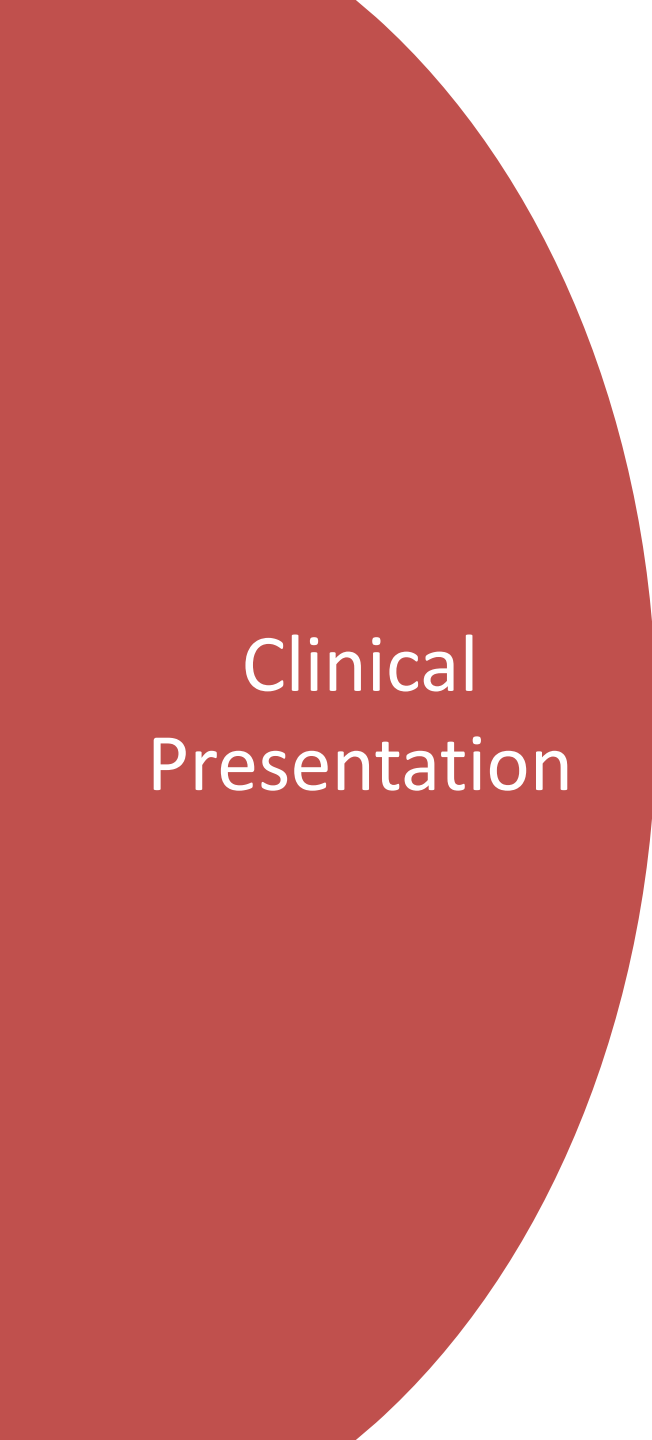
Definition

- Osteoarthritis (OA) is a common, slowly progressive disorder affecting weight-bearing diarthrodial joints.
- Characterized by progressive deterioration and loss of articular cartilage, osteophyte formation, pain, limitation of motion, deformity, and disability.
- Inflammation may or may not be present.



Pathophysiology

- Primary (idiopathic) OA: no known cause.
 - Secondary OA: linked to trauma, inflammatory arthritis, metabolic or endocrine disorders.
 - Cartilage damage triggers chondrocyte activation → matrix metalloproteinases (MMPs) enzyme release → cartilage destruction.
 - Subchondral bone remodeling and osteophyte formation.
 - Leads to joint space narrowing and pain.
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Clinical Presentation

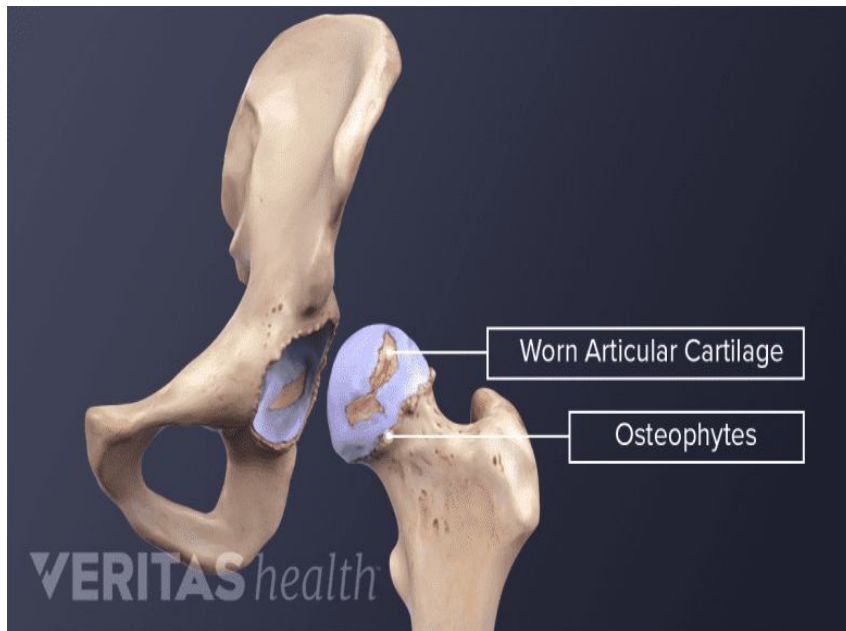
- Joint pain and stiffness (<30 min morning stiffness).
 - Crepitus, tenderness, and limited motion.
 - Bony enlargements (Heberden's and Bouchard's nodes).
 - Joint deformity in late stages.
 - Most affected: knees, hips, hands, spine.
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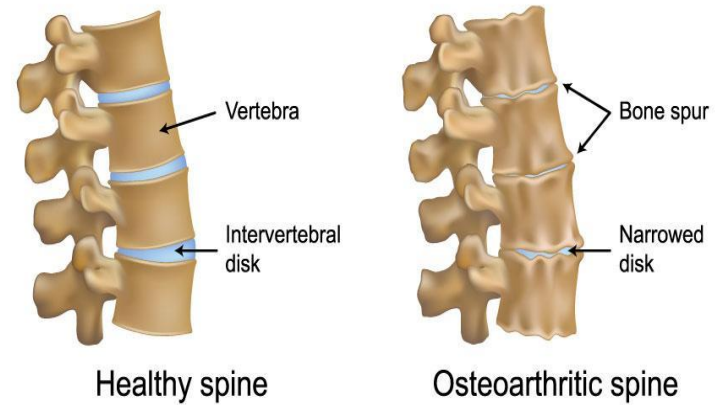
Diagnosis

- Based on history, clinical exam, and radiologic findings.
- American College of Rheumatology (ACR) criteria:
 - Hip OA: pain + radiographic osteophytes or joint space narrowing.
 - Knee OA: pain + radiographic osteophytes + age >50 or crepitus or stiffness ≤ 30 min.
- Synovial fluid: high viscosity, mild leukocytosis (<2000 cells/mm³)/non-inflammatory.





Osteoarthritis of Spine



Treatment Goals

01

Educate patients and caregivers.

02

Relieve pain and stiffness.

03

Maintain joint mobility and function.

04

Limit functional impairment.

05

Improve quality of life.

Non-Pharmacologic Therapy

Patient education
and lifestyle
modification.

Weight reduction
and dietary
counseling.

Physical therapy
(heat/cold, range of
motion exercises).

Assistive devices:
canes, braces,
insoles.

Surgical options in
severe cases.

Patient Education Goals

- **Increase understanding of OA:**
 - Explain the nature of joint degeneration, symptoms, and long-term implications.
- **Promote self-management:**
 - Encourage patients to monitor symptoms, adhere to treatment plans, and make informed decisions.
- **Reduce fear and misconceptions:**
 - Clarify that OA is manageable and not an inevitable consequence of aging.
- **Improve treatment adherence:**
 - Support consistent use of medications, physical therapy, and lifestyle changes.

Lifestyle Modification Strategies

- **1. Exercise & Physical Activity**
 - **Low-impact aerobic exercises** (e.g., walking, swimming, cycling) improve joint mobility and cardiovascular health.
 - **Strength training** enhances muscle support around joints, reducing stress and pain.
 - **Flexibility exercises** like yoga or stretching maintain range of motion.
 - *Tip:* Tailor programs to individual capacity and joint involvement.

Lifestyle Modification Strategies...

- **2. Weight Management**

- **Weight loss reduces joint load**, especially in knees and hips.
- Combine **nutritional counseling** with physical activity for sustainable results.
- Even a **5–10% reduction in body weight** can significantly improve symptoms.

Lifestyle Modification Strategies...

- Why Weight Reduction Matters in Osteoarthritis
 - **Every 1 kg of weight loss reduces knee joint load by 4 kg** during daily activities.
 - Excess weight accelerates **cartilage breakdown**, increases inflammation, and worsens pain.
 - Weight loss improves **mobility, function, and quality of life**, and may delay the need for surgery.

Lifestyle Modification Strategies...

- **3. Joint Protection Techniques**

- Use **assistive devices** (e.g., canes, braces) to reduce strain.
- Modify daily activities to avoid repetitive stress (e.g., using ergonomic tools).
- Practice **proper body mechanics** during lifting or bending.

Lifestyle Modification Strategies...

Assistive devices: canes, braces, insoles.



Lifestyle Modification Strategies...

- **4. Dietary Adjustments**

- Emphasize **anti-inflammatory foods**: omega-3 fatty acids, fruits, vegetables, whole grains.
- Limit **processed foods and sugars** that may exacerbate inflammation.
- Ensure adequate **vitamin D and calcium** intake for bone health.

Lifestyle Modification Strategies...

- **5. Pain Management & Relaxation**
 - Teach **non-pharmacologic techniques**: heat/cold therapy, massage, mindfulness, and breathing exercises.
 - Encourage participation in **support groups** or counseling for emotional well-being.

Lifestyle Modification Strategies...

6. Medication Literacy

- Educate on **safe use of NSAIDs**, topical agents, and supplements.
- Discuss potential side effects and interactions.
- Reinforce **adherence to prescribed regimens** and regular follow-up.

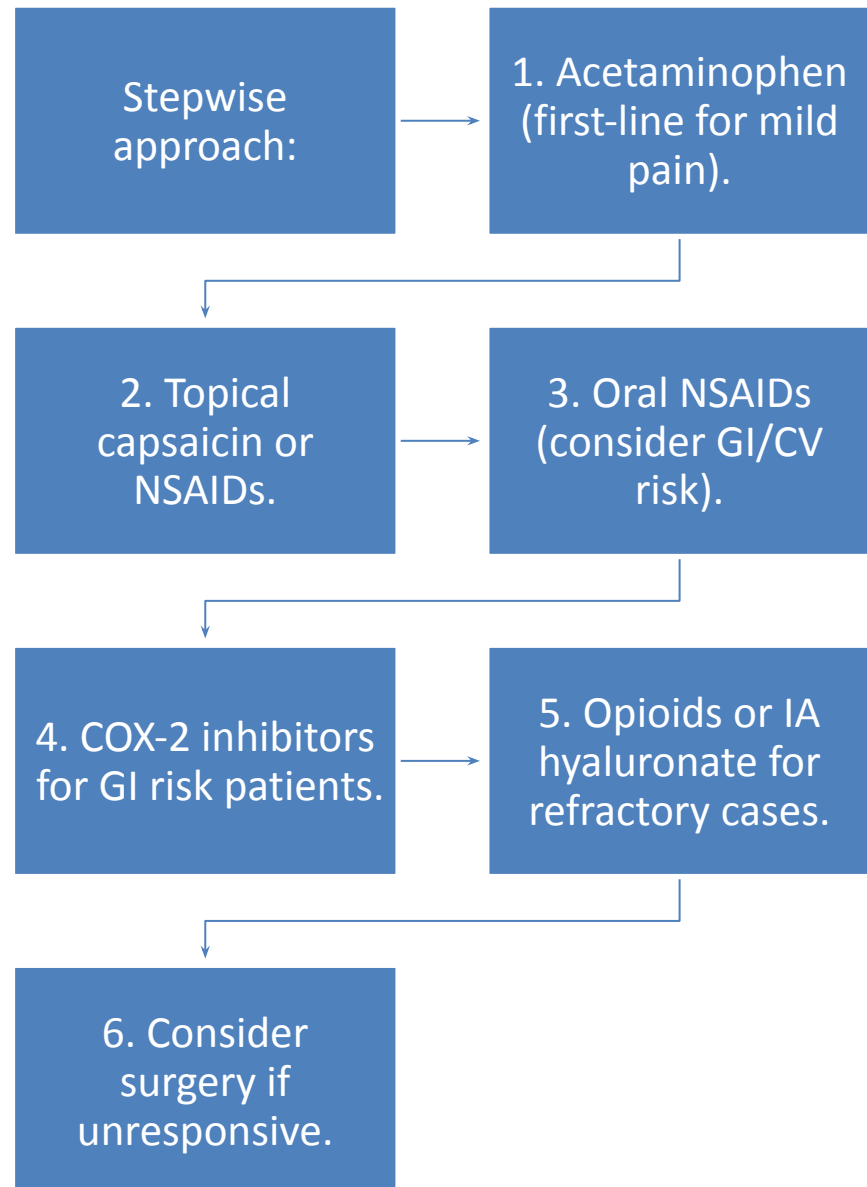
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Pharmacologic Therapy



Acetaminophen

- Why Acetaminophen Is First-Line for Mild OA Pain
 - **Recommended for mild to moderate pain:** Particularly useful in early-stage OA or when inflammation is minimal.
 - **Safer for patients with gastrointestinal risks:** Unlike NSAIDs, acetaminophen does not irritate the stomach lining or increase bleeding risk.
 - **Preferred in patients with cardiovascular or renal concerns:** It avoids the systemic effects associated with NSAIDs

Acetaminophen: Clinical Considerations

- **Dosage:** Up to 3,000–4,000 mg/day in divided doses (depending on local guidelines and patient factors). Always consider liver function.
- **Contraindications:**
 - Chronic liver disease
 - Excessive alcohol use
- **Monitoring:** Educate patients to avoid combining with other acetaminophen-containing products to prevent overdose.

Acetaminophen: Limitations

- **Less effective for inflammatory pain:** May not provide sufficient relief in advanced OA or during flare-ups.
- **No anti-inflammatory action:** Unlike NSAIDs, it does not reduce joint inflammation.

Topical Capsaicin

- **Mechanism:**

- Derived from chili peppers, capsaicin works by **depleting substance P**, a neuropeptide involved in transmitting pain signals.
- Repeated application leads to **desensitization of pain receptors** in the skin.

- **Use:**

- Applied 3–4 times daily to affected joints
- Onset of relief may take **1–2 weeks** of consistent use

Topical Capsaicin...

- **Benefits:**

- Effective for **superficial joints** (e.g., hands, knees)
- Minimal systemic absorption
- Useful in patients who cannot tolerate NSAIDs

- **Cautions:**

- May cause **burning or stinging** sensation initially
- Avoid contact with eyes or mucous membranes
- Wash hands thoroughly after application (or use gloves)

Oral NSAIDs (consider GI/CV risk).

- Role of Oral NSAIDs in OA Management
 - **Mechanism:** Inhibit cyclooxygenase (COX) enzymes, reducing prostaglandin synthesis and inflammation
 - **Indication:** Moderate to severe OA pain, especially when topical agents or acetaminophen are insufficient
 - **Common agents:** Ibuprofen, naproxen, diclofenac, celecoxib

Common NSAIDs and Their Dosing

NSAID	Typical Adult Dose	Max Daily Dose
Ibuprofen	400–800 mg every 6–8 hours	3,200 mg (Rx); 1,200 mg (OTC)
Naproxen	250–500 mg twice daily	1,000 mg
Diclofenac	50 mg 2–3 times daily (oral)	150 mg
Celecoxib	100–200 mg once or twice daily	400 mg

Note: Dosing may vary based on age, renal function, and comorbidities. Always consult local guidelines.

COX-1 vs COX-2 Inhibitors

Feature	COX-1 Inhibitors	COX-2 Inhibitors
Enzyme Target	Cyclooxygenase-1 (constitutive)	Cyclooxygenase-2 (inducible)
Physiologic Role	Protects GI mucosa, supports platelets	Mediates inflammation and pain
Common Drugs	Ibuprofen, naproxen, aspirin	Celecoxib, etoricoxib, meloxicam (partially selective)
GI Risk	Higher (ulcers, bleeding)	Lower
CV Risk	Lower (e.g., naproxen)	Higher (especially rofecoxib, withdrawn)
Platelet Effect	Inhibits platelet aggregation	Minimal effect

Opioids or IA hyaluronate for refractory cases.

- In refractory OA cases, **IA hyaluronate injections are preferred** over opioids due to better safety and joint-specific benefits, though **both have limited long-term efficacy**.
- Opioids are reserved for select patients with severe pain unresponsive to other therapies.

Intra-Articular Hyaluronate (Viscosupplementation)

- **Mechanism:**

- Mimics natural synovial fluid to improve joint lubrication and shock absorption
- May reduce pain and improve mobility in **knee OA**, especially in early to moderate stages

- **Evidence & Use:**

- A 2025 umbrella review found **modest benefits** in pain reduction and function, particularly in patients with mild to moderate knee OA
- Common brands: Euflexxa, Synvisc, Durolane
- Typically administered as **1–3 weekly injections**, with effects lasting up to 6 months

Intra-Articular Hyaluronate (Viscosupplementation)...

- **Advantages:**
 - **Localized treatment** with minimal systemic side effects
 - Suitable for patients with **GI, CV, or renal risks** who cannot tolerate NSAIDs
- **Limitations:**
 - **Variable response** among patients
 - **Not recommended for hip OA** due to deeper joint anatomy
 - Cost and access may be barriers in some settings

Opioids

- **Mechanism:**

- Bind to central opioid receptors to block pain perception
- Used for **severe, refractory OA pain** when other treatments fail

- **Evidence & Use:**

- Studies show **limited long-term benefit** and **faster OA progression** among opioid users
- Common agents: tramadol (weak opioid), oxycodone, morphine
- Often considered a **last resort** before surgical intervention

- **Risks:**

- **Dependence, tolerance, and withdrawal**
- **Cognitive impairment, constipation, falls**, especially in older adults
- CDC and rheumatology guidelines advise **cautious prescribing**

Clinical Decision

Rx	Best For	Risks	Duration of Relief
IA Hyaluronate	Mild–moderate knee OA	Injection site pain, cost	Up to 6 months
Opioids	Severe, refractory OA pain	Dependence, sedation, falls	Short-term

Surgical interventions

- When Is Surgery Recommended
 - Persistent pain despite medications, physical therapy, and injections Severe joint damage confirmed by imaging
 - Functional limitations affecting daily life or work
 - Failure of less invasive interventions (e.g., IA hyaluronate, bracing)

Surgical intervention options

- **1. Joint Replacement (Arthroplasty)**
 - **Most common for advanced hip and knee OA**
 - Damaged joint surfaces are replaced with prosthetic components
 - **Benefits:** Significant pain relief, improved mobility, long-term durability (10–20 years)
 - **Risks:** Infection, blood clots, prosthesis loosening, need for revision surgery

Surgical intervention options...

- **2. Osteotomy**

- Realigns bones to shift weight away from damaged joint areas
- Often used in **younger patients with knee OA** and localized cartilage loss
- **Benefits:** Preserves native joint, delays need for replacement
- **Risks:** Longer recovery, less predictable outcomes than arthroplasty

Surgical intervention options...

- **3. Joint Fusion (Arthrodesis)**
 - Bones are fused to eliminate painful movement
 - Common in **small joints** (e.g., spine, fingers, ankle)
 - **Benefits:** Pain relief and stability
 - **Risks:** Loss of joint mobility, adjacent joint stress

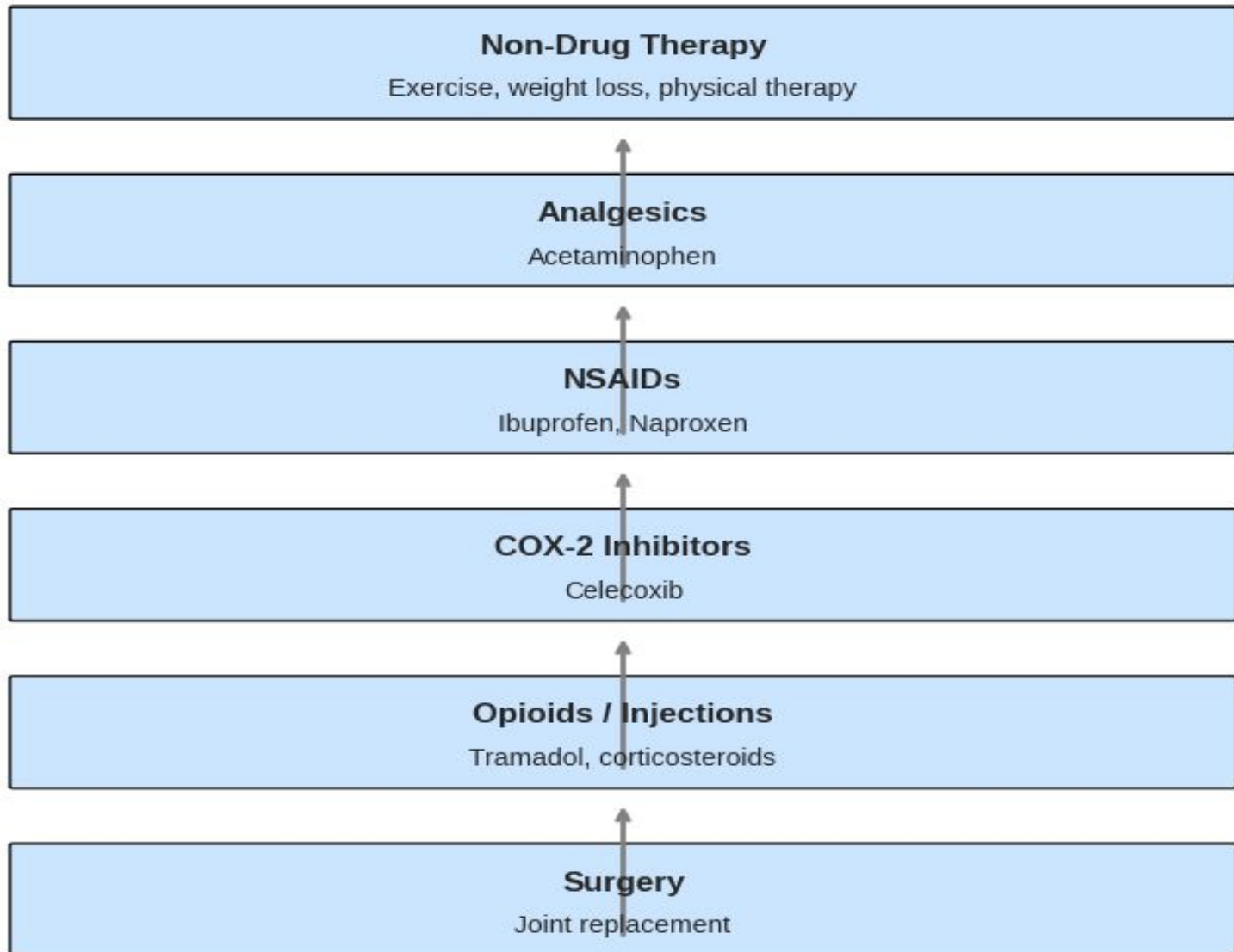
Surgical intervention options...

- **4. Debridement or Synovectomy**
 - Removal of damaged cartilage or inflamed synovial tissue
 - Typically done arthroscopically in early OA stages
 - **Benefits:** Temporary symptom relief
 - **Risks:** Limited long-term efficacy

Pre- and Post surgical Operation Considerations

- **Pre-op:** Optimize comorbidities (e.g., diabetes, hypertension), assess surgical risk, patient education
- **Post-op:** Physical rehabilitation, infection prevention, pain management, long-term follow-up

OA management flowchart



Summary

Drug Class	Examples	Typical Dosage	Route	Notes
NSAIDs	Ibuprofen, Naproxen, Diclofenac	Ibuprofen: 400–800 mg TID/QID	Oral, Topical	Effective for pain/inflammation; monitor GI, renal, and CV risks
COX-2 Inhibitors	Celecoxib	100–200 mg once or twice daily	Oral	Lower GI risk than traditional NSAIDs; caution in CV disease
Analgesics	Acetaminophen	500–1000 mg QID (max 4g/day)	Oral	First-line for mild OA; less effective for inflammation
Topical Agents	Diclofenac gel, Capsaicin	Diclofenac: apply QID; Capsaicin: TID–QID	Topical	Useful for localized OA (e.g., hands, knees); fewer systemic effects
Intra-articular Steroids	Triamcinolone, Methylprednisolone	10–40 mg per injection	Injection	Short-term relief; limit to 3–4 injections/year per joint
Hyaluronic Acid	Hylan G-F 20, Sodium hyaluronate	1 injection/week for 3–5 weeks	Injection	Mixed evidence; used for knee OA
Duloxetine	Cymbalta	30–60 mg once daily	Oral	For chronic pain and OA with central sensitization
Opioids	Tramadol, Codeine	Tramadol: 50–100 mg QID (max 400 mg/day)	Oral	Reserved for severe pain; risk of dependence and side effects

References

- Pharmacotherapy Handbook, 7th Edition – Chapter 2: Osteoarthritis.
- American College of Rheumatology (ACR) Guidelines.
- Recent literature on OA pharmacologic management.